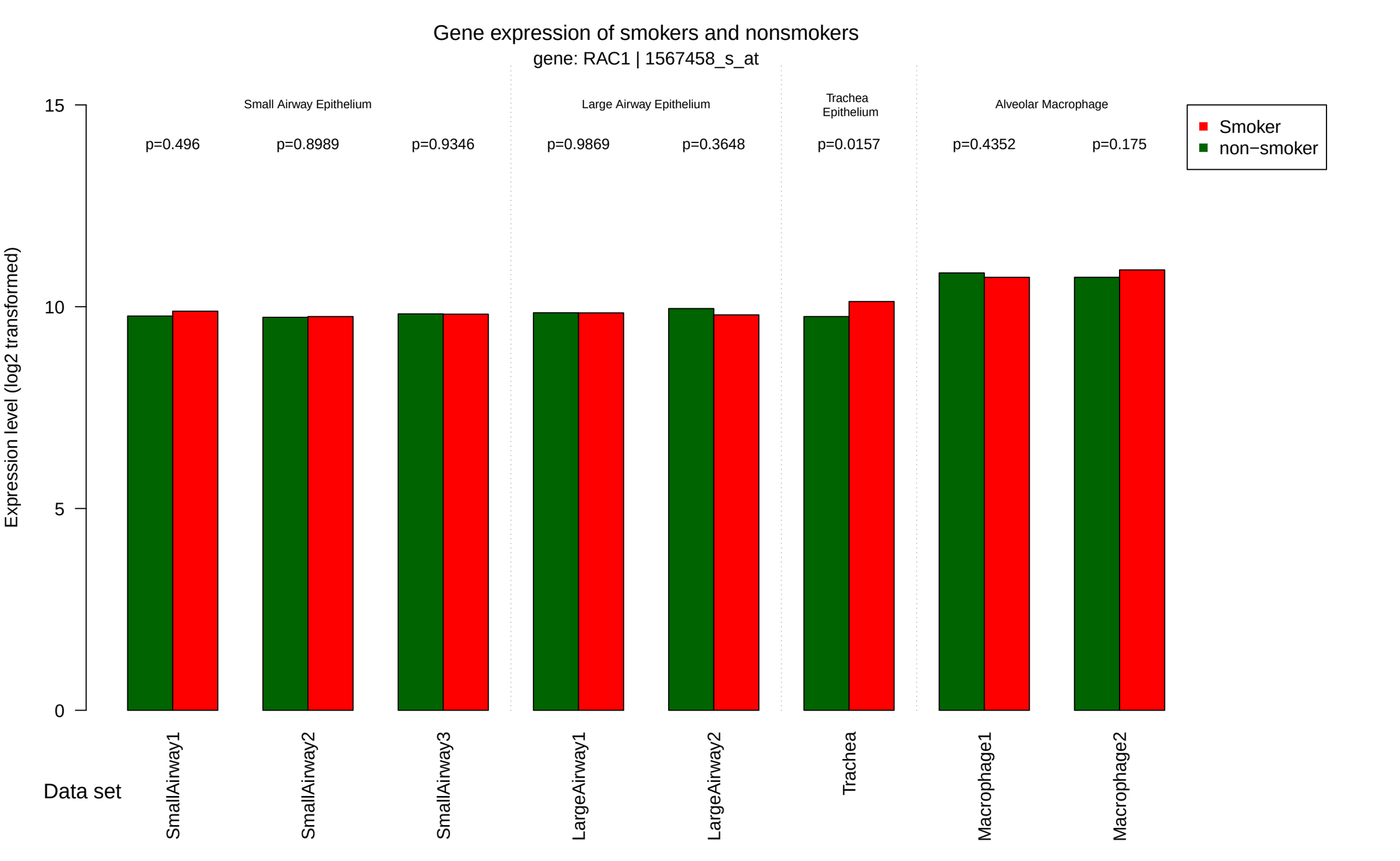
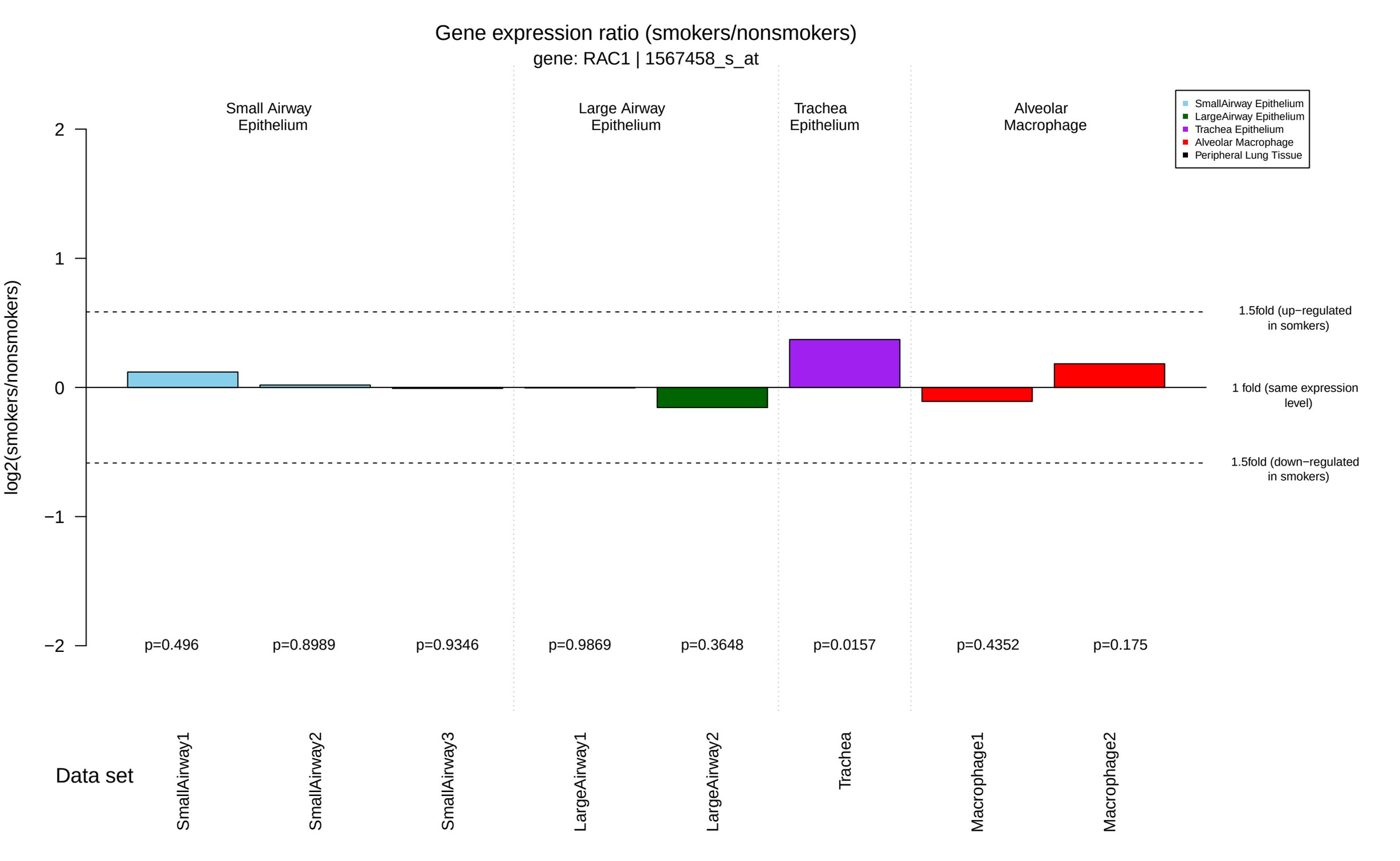
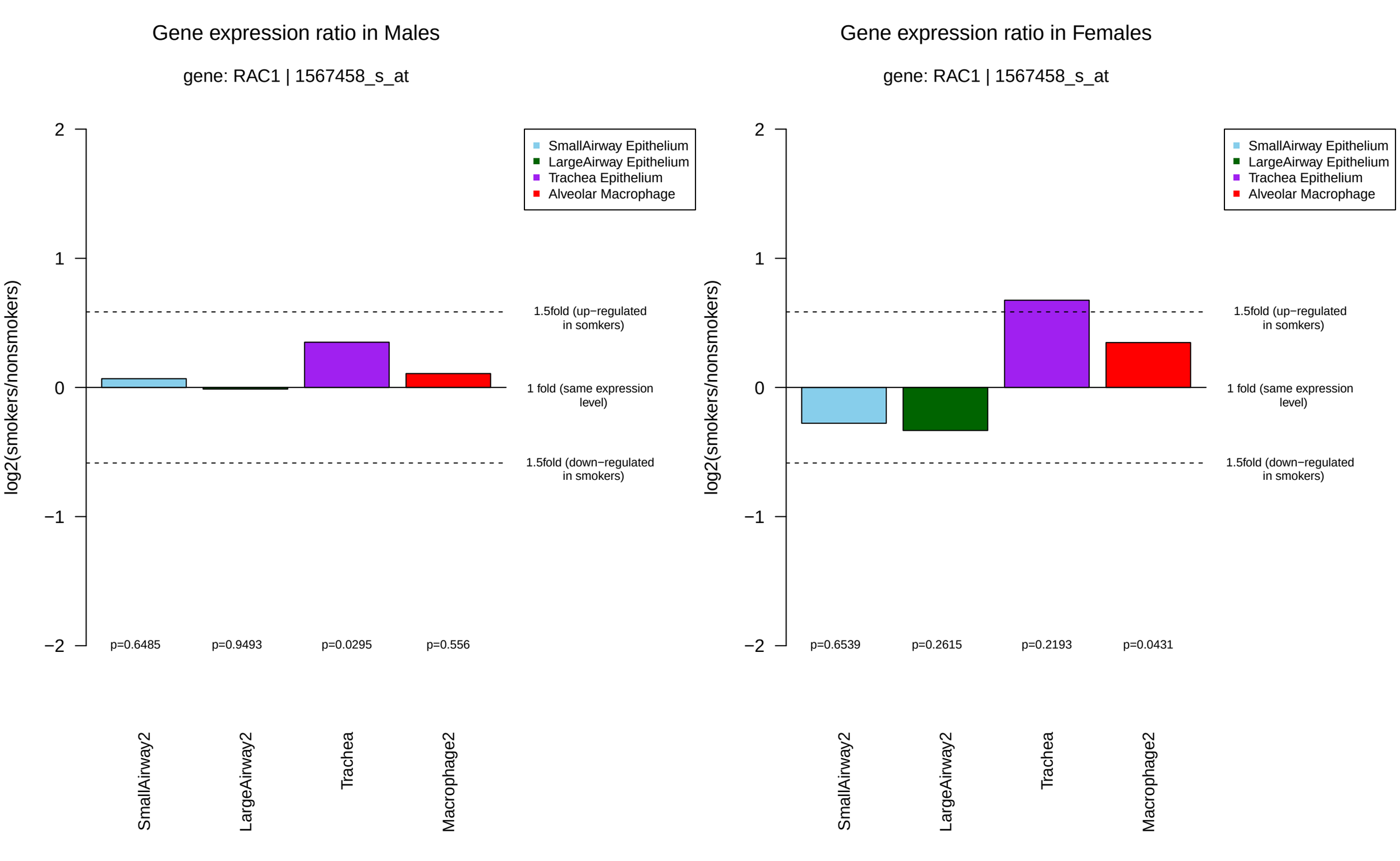
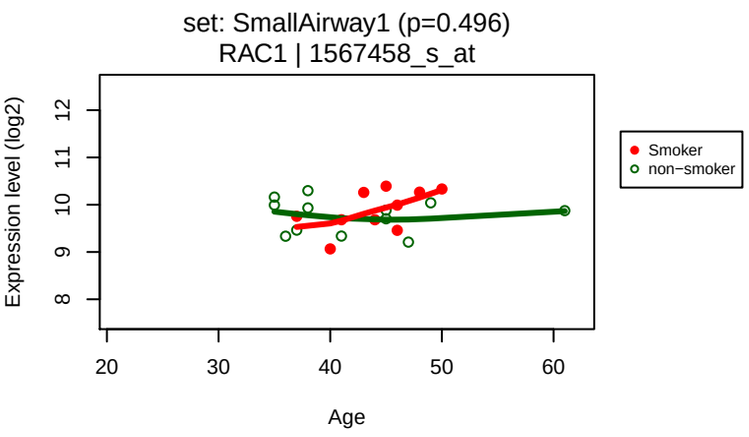
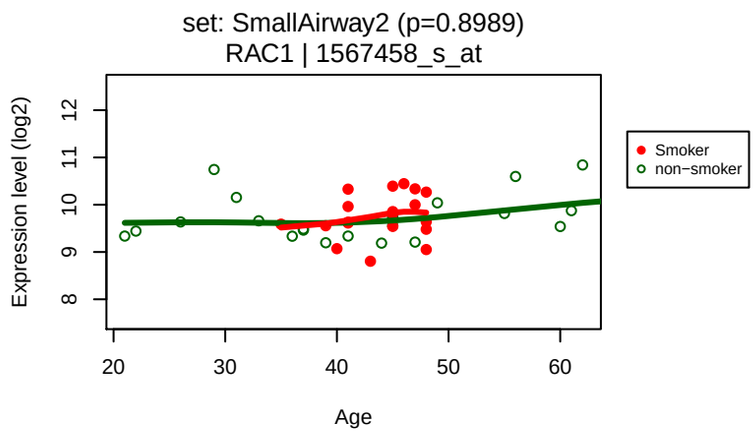
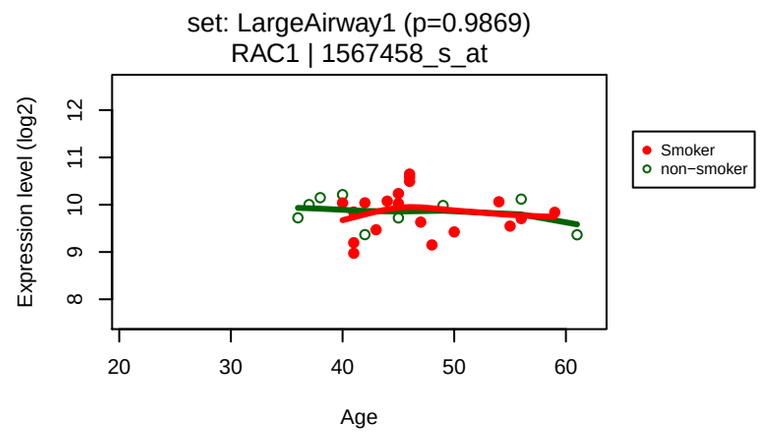

Gene expression of smokers and nonsmokers
gene: RAC1 | 1567458_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.496
p=0.8989
p=0.9346
p=0.9869
p=0.3648
p=0.0157
p=0.4352
p=0.175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

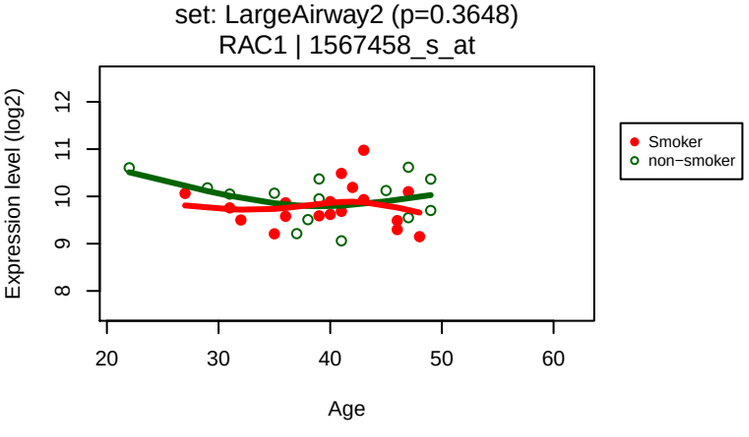

Gene expression ratio (smokers/nonsmokers)
gene: RAC1 | 1567458_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.496
p=0.8989
p=0.9346
p=0.9869
p=0.3648
p=0.0157
p=0.4352
p=0.175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

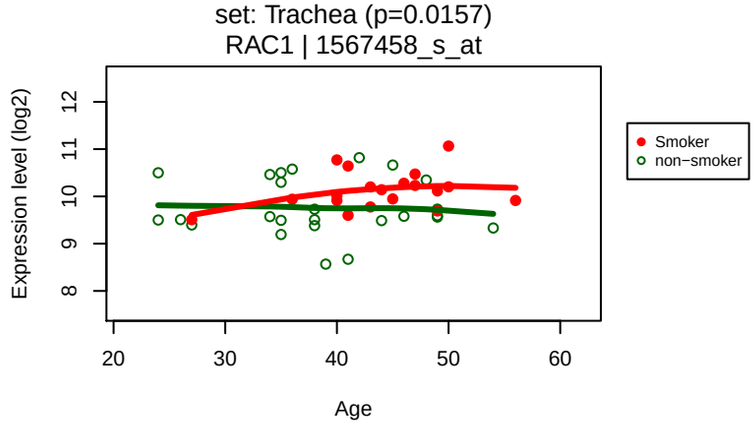

Gene expression ratio in Males
gene: RAC1 | 1567458_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6485
p=0.9493
p=0.0295
p=0.556
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC1 | 1567458_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6539
p=0.2615
p=0.2193
p=0.0431
SmallAirway2
LargeAirway2
Trachea
Macrophage2

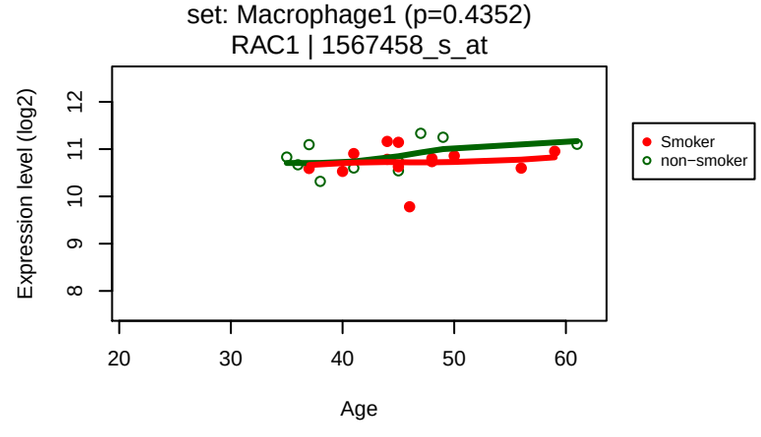

set: SmallAirway1 (p=0.496)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker

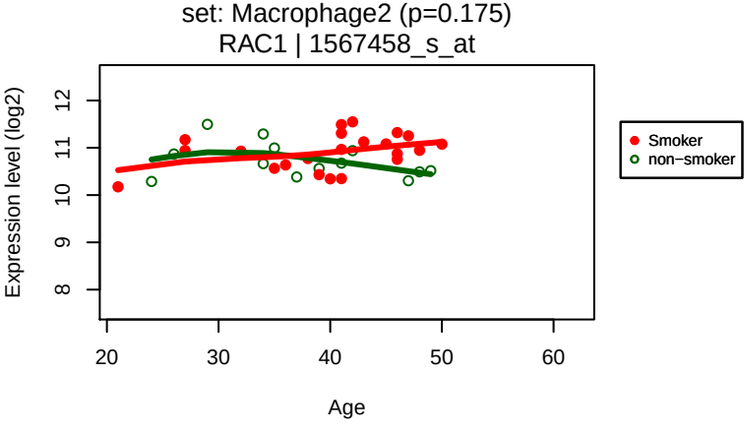

set: SmallAirway2 (p=0.8989)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker

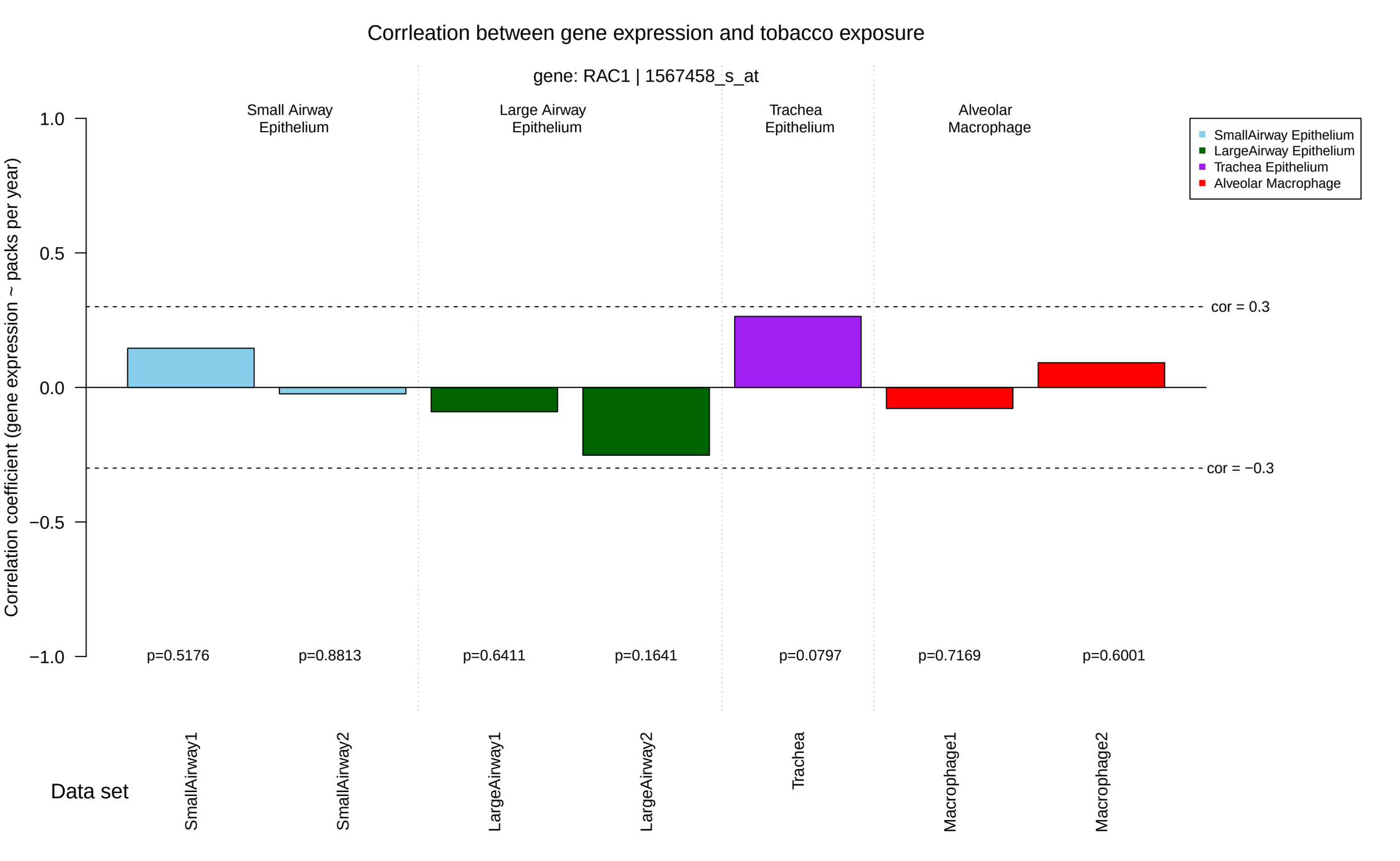

set: LargeAirway1 (p=0.9869)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3648)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0157)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4352)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.175)
RAC1 | 1567458_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC1 | 1567458_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5176
p=0.8813
p=0.6411
p=0.1641
p=0.0797
p=0.7169
p=0.6001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2